

| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|-----|------|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |     | (iv) |  | <b>One (x1) from:</b><br>raise price/excise duty/tax (1)<br>education on dangers(1)<br>encourage use of alternatives (1)<br>raise age for buying cigarettes (1)<br>plain packaging (1)<br>Remove from display (1)<br>NOT ban smoking                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                 | 1   |     | 1     |       |      |
| 5        | (c) |      |  | <i>Indicative content</i> <ul style="list-style-type: none"> <li>• two groups/ smokers and non smokers</li> <li>• measure breathing rate</li> <li>• repeat</li> <li>• find the mean</li> <li>• compare the two groups</li> <li>• <b>two</b> fair testing –</li> <li>• e.g. same age/ fitness/ gender/ length of time smoking/ number smoked per day/ whether they had exercised prior to test</li> </ul><br><b>5 – 6 marks</b> : Detailed description of the entire investigation<br><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> |                 |     | 6   | 6     |       | 6    |